# Large Convolution Kernel Network With Edge Self-Attention for Oriented SAR Ship Detection

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***Abstract*—Synthetic aperture radar (SAR) ship detection is a popular area in remote sensing, which has broad applications in fishery management, maritime rescue and marine detection. In recent years, deep learning has been successfully applied in this field, but SAR ship detection still faces some challenges, such as speckle noise, strong scattering interference from complex backgrounds, and insufficient use of SAR image characteristics. These issues result in inaccurate ship positioning in complex scenes and a high rate of false alarms. To address the above challenges, we propose a large convolution kernel network with edge self-attention for oriented SAR ship detection (LKE-Det), which can accurately detect arbitrarily rotated ship targets in SAR images. First, we propose a backbone network based on large convolutional kernels, which has a large receptive field and can enhance the long-range dependencies of ships. The backbone adopts a bifurcated gate unit structure to combine the multiscale parallel convolution kernel module and the large convolution kernel attention module. The structure captures feature information of different scales while obtaining sufficient context information. Second, we adopt an edge self-attention mechanism, which uses an edge detection algorithm to extract the gradient matrix of the image and integrate it into the feature pyramid structure to effectively enhance the contour information of ships. The extensive experiments on SSDD and RSDD-SAR datasets demonstrate that our LKE-Det outperforms the state-of-the-art methods.**

***Index Terms*—Deep learning, edge self-attention, large convolution kernel, oriented ship detection, synthetic aperture radar (SAR).**

## I. Introduction

SYNTHETIC Aperture Radar (SAR) is a high-resolution microwave imaging radar that can obtain high-resolution radar images under complex climate conditions without being affected by weather and other factors [1]. SAR image ship detection is one of the more important research directions in target detection based on SAR images and is of great significance in maritime merchant ship supervision and military activity monitoring [2].

The existing techniques for ship detection in SAR images mainly include traditional Constant False Alarm Rate Detector (CFAR)-based detection and deep learning-based detection [3]. The detection algorithms based on CFAR dynamically adjust the detection threshold according to the statistical characteristics of the radar echo data to reduce the radar false alarm rate [4]. Such classic method is also widely used in SAR image ship target detection. He et al. [5] proposed a method to automatically detect ships using three superpixel-level dissimilarity measures. Lin et al. [6] proposed a superpixel-level Fisher vector to describe the difference between target and clutter. Liu et al. [7] proposed a simplified polarization SAR ship detection method based on K-Wishart distribution. The false alarm rate is obtained by bisection method to set the CFAR threshold, realizing adaptive ship detection in complex ocean background. However, traditional SAR ship detection methods only extract simple underlying features, which are insufficient for recognizing scenarios with small differences such as near shore.

Recently, deep learning methods with automatic feature extraction are becoming increasingly popular for object detection [8]. In the field of optical image object detection, many advanced algorithms have been proposed. These algorithms can be roughly divided into two categories: 1) one-stage algorithms YOLO V8 [9], Centernet [10], YOLO V9 [11], YOLO V10 [12]; and 2) two-stage algorithms Faster R-CNN [13], Cascade R-CNN [14], SPP-Net [15]. Following the above studies, many CNN-based SAR ship detection algorithms have been proposed [16], [17], [18], [19], [20]. The neural network model can be driven by a large number of annotated datasets to realize automatic learning of target detection features, which is able to extract richer features compared with traditional algorithms. Gao et al. [21] proposed an anchor-free convolution neural network based on dense attention feature aggregation for ship detection in SAR images. This method uses a lightweight feature extractor to extract multiscale ship features and uses depth-wise separable convolution to reduce network parameters and improve detection speed. Yang et al. [22] introduced a robust one-stage ship detector featuring a coordinate attention module and a receptive field expansion module. This method can effectively alleviate complex background interference and capture multiscale contextual information. Fu et al. [23] proposed a feature balancing and refinement network to eliminate the influence of anchor

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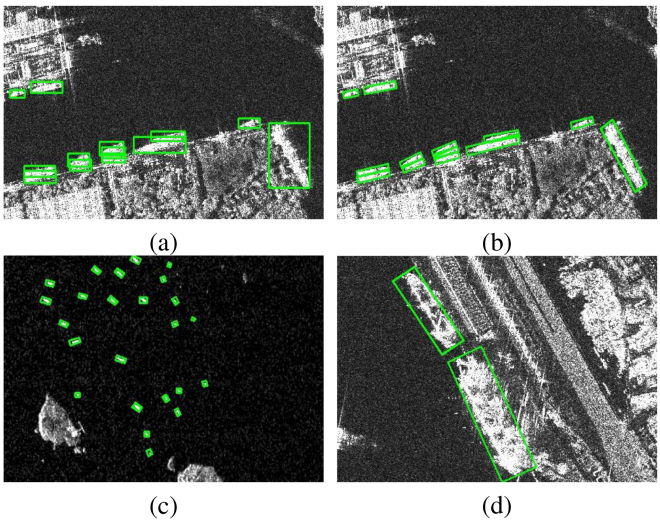



Fig. 1. Some typical scenes of densely aligned, arbitrarily oriented, multiscale in SAR images. (a) (b) (d) ships densely arranged scenes, (c) the scene with many small ships.

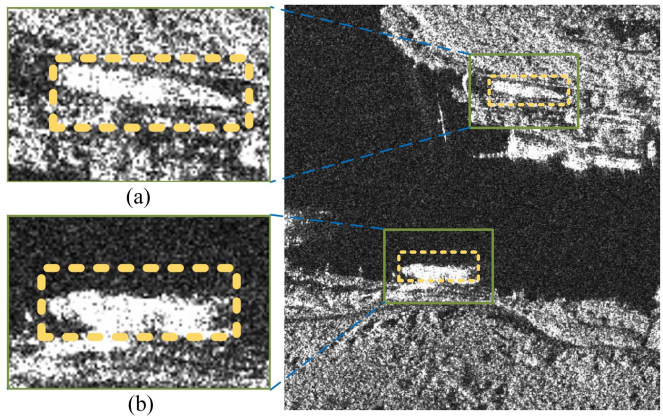



Fig. 2. Comparison of surrounding feature information of positive and negative samples with similar local features. (a) the negative sample, (b) the positive sample.

points by adopting a general anchor-free strategy of directly learning to encode bounding boxes. Although these methods have made great progress, they primarily rely on horizontal bounding boxes (HBB). As shown in Fig. 1(a), strategies based on HBB make it difficult to accurately locate the ships. SAR uses a high-altitude imaging mode, in which the ship target has a large aspect ratio and an arbitrary orientation. Traditional horizontal bounding boxes cannot accurately reflect the target's true shape. In addition, there is a large amount of coherent speckle noise in SAR images. The horizontal bounding box introduces a lot of background interference, which is not conducive to the precise positioning of the target. However, as shown in Fig. 1(b), the method based on oriented bounding box (OBB) can effectively avoid the above problems. Therefore, scholars have developed many OBB detection algorithms in the field of optical remote sensing images, including RoITransformer [24], S2anet [25], R3Det [26], ReDet [27], and Oriented R-CNN [28]. Based on these methods, many SAR ship detection algorithms have been developed to achieve more accurate detection results [29], [30], [31], [32], [33].

OBB can better locate the target and more closely conforms to the shape of the ship target [34]. OBB detection methods are increasingly proposed for target detection in SAR images. Yue et al. [35] proposed a two-stage anchor free detector that utilizes a dynamic rotation positive sample mining module specifically designed for hard positive sample mining in directional ship detection. Ju et al. [36] proposed a new five parameter polar coordinate density regression detector that uses only one centroid, one mapped polar diameter, and two polar angles to handle the overlapping problem of HBB and the boundary discontinuity problem of OBB. Gao et al. [37] utilized elliptical encoding to effectively utilize the geometric and scattering characteristics of ship targets. By combining the pitch interval with the decoding process, the negative impact of boundary discontinuity is avoided. In addition, an elliptical Gaussian distribution heatmap and a bimodal maximum pooling strategy were used to handle the separation problem of densely distributed ships in offshore scenes. However, these algorithms are insufficient in utilizing the unique global and contextual information in SAR images and in making full use of the characteristics of SAR images that are different from visible light images. Our method can effectively solve the above problems.

*Expanding the effective receptive field with large convolution kernels:* Most SAR ship detection algorithms focus primarily on the local information within images, making it challenging to model the global information of the images effectively. The local connectivity of CNN convolution kernels limits the model's ability to learn global ship information, making detection vulnerable to interference from complex backgrounds. Unlike visible light images, the available feature information of ship targets in SAR images is very limited. Moreover, as shown in Fig. 1(d), buildings on land also exhibit significant scattering characteristics and often resemble ships in shape. Consequently, the lack of extensive contextual information leads to poor detection results. As shown in Fig. 2, it is difficult to determine whether the buildings on land that are similar in shape to ships are real targets or false alarms solely by relying on the local features within the yellow marked boxes. However, as can be seen from the figure, the surrounding information of the ships in the green boxes can greatly help determine whether it is a false target. Moreover, networks with large convolution kernels can extract features over a larger range, making them more robust to local noise and helping the model perform better in noisy SAR images.

*Make full use of the strong scattering characteristics of ships in SAR images:* Ships are typically constructed from metallic materials, which exhibit strong scattering characteristics when illuminated by radar waves [38]. The significant differences in both material and shape between ships and their surrounding environment cause noticeable variations in reflectivity. As shown in Fig. 1, whether the ships are at sea or on shore, their edges will show a large reflectivity change with the surrounding sea surface. Particularly on the sea surface, where the smooth water generally reflects minimal radar waves, ships generate strong echoes,

creating distinct edge features. Although buildings on shore also exhibit strong scattering, they lack the distinct gradient changes seen around ships. Therefore, our network integrates this gradient change with other features to improve detection accuracy and robustness.

Leveraging these unique SAR image features can significantly improve the accuracy of ship detection. Based on the above analysis, we propose a novel method called large convolution kernel network with edge self-attention for oriented SAR ship detection (LKE-Det). The contributions of this article are summarized as follows.

1) We proposed a large receptive field backbone network, which combines a multiscale parallel convolution module with a large convolution kernel attention mechanism using a bifurcated gate unit structure. It can capture feature information of different scales and obtain sufficient contextual information.
2) In order to make full use of the obvious gradient changes around the ship in SAR images, this article proposes an edge self-attention mechanism. The structure integrates frequency domain information of ships into the feature pyramid structure, which can significantly improve the network's detection capabilities in nearshore scenes.
3) Extensive experiments on SSDD datasets and RSDD-SAR datasets demonstrate the effectiveness of the proposed network. The proposed method more advanced detection performance compared to many other arbitrarily oriented object detectors and advanced SAR detection algorithms.

The rest of this article is organized as follows. First, Section II provides a detailed introduction to the proposed method. Second, Section III describes the experimental implementation details such as the adopted data sets and evaluation metrics. Then, Section IV shows some ablation experiments and comparison experiments with other methods. Finally, Section V concludes this article.

## II. Methods

In this section, we initially introduce the overall architecture of the proposed method. Then we describe the detailed structure of the backbone network, which includes the multiscale parallel convolution kernel (MPK) module, the large convolution kernel attention (LKA) module, and Feed Forward Network (FFN) module. Finally, we introduce the architecture of the Edge self-attention (Edge-FPN) mechanism based on image gradient information.

### A. Overall Architecture

This detector is a two-stage, anchor-based network called LKE-Det, consisting of three components: 1) the backbone with large convolution kernels for feature extraction; 2) the Edge-FPN for enhancement of targets and suppression of background clutter; and 3) the detection head based on Oriented-RCNN for classification and regression. LKE-Det can utilize the extensive contextual information of ships and the unique properties of SAR imaging mechanism to improve ship detection performance in complex scenes. Multiscale feature maps obtained through the backbone network from the input image, which are combined with the frequency domain information of the image and then inputted into the detection head. The backbone of LKE-Det and the Edge self-attention module are illustrated in Fig. 3. The backbone network consists of four stages, each of which contains multiple LK blocks, forming a multiscale hierarchical framework. Simultaneously, as shown in Fig. 4, each LK-block consists of three modules: 1) MPK module; 2) LKA module; and 3) FFN module. In addition, since the pixel values in SAR images reflect the scattering characteristics of objects, there exists a distinct gradient change between the ship and the surrounding environment. Based on this innovative idea, the edge self-attention mechanism incorporates the frequency domain information of the image into the FPN to enhance the contrast between the ship and the complex background. A detailed structural explanation will be provided in the following sections.

### B. Large Kernel Block

The model's backbone is segmented into four stages connected by downsampling blocks. Specifically, the first downsampling module uses a stride of 4 and a $7 \times 7$ convolutional layer to convert the raw input into C-channel feature maps. In the remaining three stages, each stage employs a stride of 2 and a $3 \times 3$ convolutional layer, achieving downsampling operation so that the number of channels in the four stages are $\{C, 2C, 4C, 8C\}$, respectively. In addition, each stage comprises multiple LK-blocks, where each block consists of three submodules. The numbers of such blocks in the four stages are denoted by N = (N1, N2, N3, N4), and the specific experimental parameters are determined by Section IV. As illustrated in Fig. 4, the main structure of backbone is consist of a MPK module, a LKA module, and a FNN module in each basic block. The MPK module is utilized to capture multiscale texture features and contextual information. Within the LKA module, large receptive field convolution kernels are employed instead of regular convolution kernels to capture long-range dependencies and location information in the image. The MPK module and the LKA module are combined through a bifurcation gate unit to facilitate the extraction and interaction of spatial feature information. The outputs of the two branches are fused through element-wise multiplication and Conv1 $\times$ 1 convolution for channel information integration. This structure effectively extracts representative spatial information from the image, covering both local and global dependencies. The FFN structure is serialized after the gate unit to achieve channel mixing and feature refinement.

### C. MPK Module

The MPK module proposed in this article solves the problem of large scale difference of ships in SAR images by effectively capturing multiscale texture features. The primary function of this module is to extract feature information at multiple scales while also capturing contextual details from local features. Remote context information is managed by our proposed LKA modules. In order to prevent too much noise, we do not use too large convolution kernel in the MPK module.

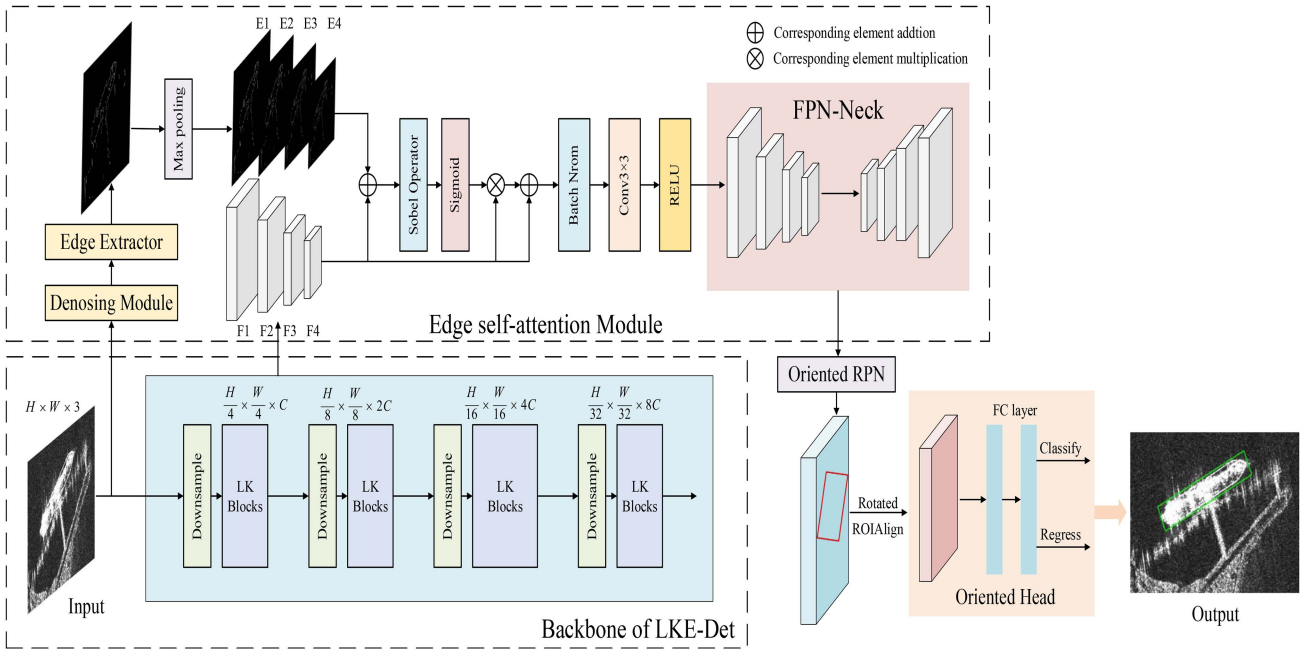



Fig. 3. Main structure of proposed network. The backbone of LKE-Det consists of four stages, each of which comprises multiple LK-Blocks. The detailed representation of LK-Blocks is illustrated in Fig. 4. The outputs of each stage of the backbone are denoted as [F1, F2, F3, F4]. Following the processing of the input image by Denoising and Edge Extractor, [E1, E2, E3, E4] is obtained, containing the gradient information of the image and matching the size of the feature map extracted by the backbone. The feature map generated by the backbone and the gradient matrix of the original image are fused to achieve edge self-attention.

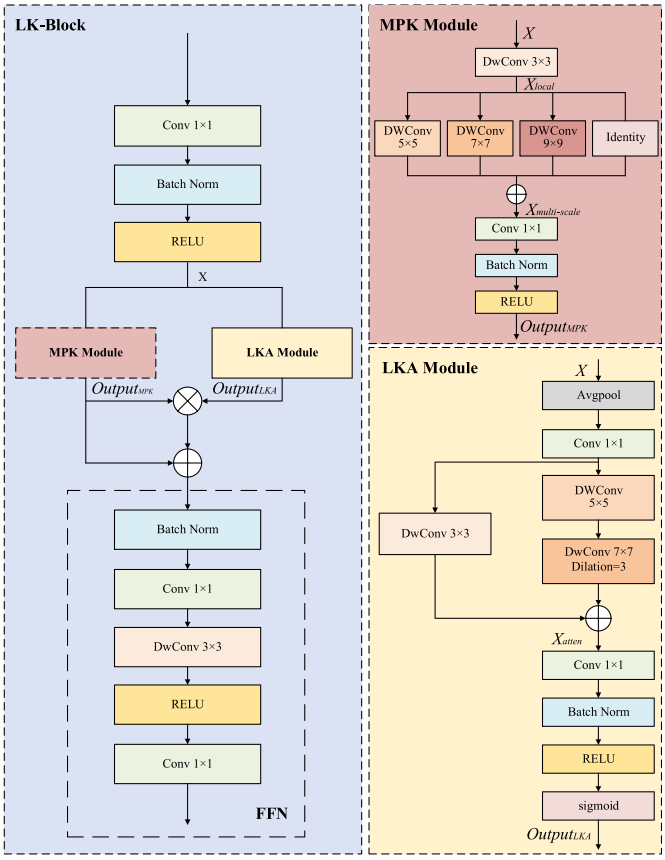



Fig. 4. Structure of the LK-Block. The Block contain a MPK module, a LKA module, and a FFN module.

The architecture of the MPK module starts with a $3 \times 3$ small kernel depth-separable convolution, responsible for capturing local information. Following this, inspired by the Unireplk [39] concept, a set of parallel depth convolutions with different kernel sizes is employed to capture multiscale local information. It is crucial to highlight that, our approach does not rely on dilated convolutions to achieve large receptive fields. Instead, we exclusively use regular convolutions. The overall mathematical expression of the MPK module's structure is as follows:

$$X_{\text{local}} = \text{DWconv}_{3\times3}(X), \tag{1}$$

$$X_{\text{multiscale}} = \sum_{n=1}^{3} \text{DWconv}_{k(n)\times k(n)}(X_{\text{local}}) + X_{\text{local}}, \tag{2}$$

$$\text{Output}_{\text{MPK}} = \text{RELU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(X_{\text{multiscale}}\right)\right)\right). \tag{3}$$

In the equations, $X$ represents the input feature, $X_{\text{local}}$ represents the local feature map obtained by applying a $3 \times 3$ depth-wise separable convolution, which corresponds to the Identity in Fig. 4. $X_{\text{multiscale}}$ represents the result of fusion of the multikernel-size deep features extracted by convolution kernels of different sizes. The final step involves applying a $1 \times 1$ convolution to achieve channel fusion of the multiple convolution results. In our experiment, we set $\text{k}_{(1,2,3)} = 5, 7, 9$. $Output_{\text{MPK}}$ denotes the output of MPK module.

This structure solves the challenges posed by scale changes and utilizes multiscale feature extraction to improve the accuracy and robustness of ship target detection in SAR images. Furthermore, the integration of the MPK module with the LKA

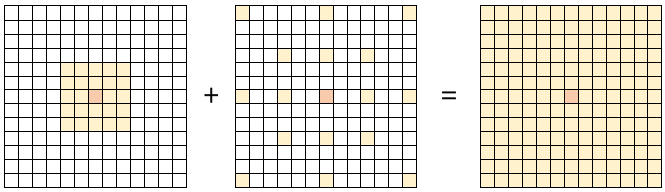


Fig. 5. Decomposition diagram of a large depth-wise convolution. The diagram demonstrates that a $13 \times 13$ depth-wise convolution can be achieved by combining a $5 \times 5$ depth-wise convolution and a $5 \times 5$ depth-wise dilated convolution with a dilation rate of 3.

module, which will be introduced in Section II-D, can capture local and remote context information.

### D. LKA Module

In order to address the challenge of ship target detection in SAR images, extensive surrounding semantic information is necessary. To effectively capture long-range contextual information, we focus on increasing the effective receptive field, enabling the network to extract more spatial structure information around the ship. Recent research [40], [41], [42] has demonstrated that the large depth-wise separable convolution can effectively increase the effective receptive field. However, directly employing it introduces a significant number of computations and network parameters, which can be computationally expensive and inefficient.

To overcome this problem, we draw inspiration from the large kernel convolution decomposition idea presented in the Visual Attention Network [43]. We decompose a K $\times$ K large convolution kernel into a combination of a depth-wise separable convolution and a depth-wise dilation convolution with dilation rate d. The decomposition details are illustrated in Fig. 5. The equivalent convolution kernel size and the actual receptive field of the dilated convolution can be calculated using the following formulas:

$$K' = d \times (k-1) + 1 \tag{4}$$

$$RF_{i+1} = RF_i + (K' - 1) \times Si \tag{5}$$

where $d$ is the rate of dilation, $k$ is the size of the dilation convolution kernel, $K'$ represents the equivalent convolution kernel size of dilation convolution, $RF_{i+1}$ represents the receptive field of the current layer, $RF_i$ represents the receptive field of the previous layer, and $Si$ represents the product of strides of all previous layers.

In our experiments, we employ a $5 \times 5$ depth-wise separable convolution followed by a $7 \times 7$ dilation convolution with an expansion coefficient of 3 to achieve an effective $23 \times 23$ large kernel convolution. In addition, we incorporate a $3 \times 3$ depth-wise convolution to preserve local features.

$$F = \text{Conv}_{1\times1}\left(P_{\text{avg}}(X)\right) \tag{6}$$

$$X_{\text{atten}} = \text{DWconv}_d\left(\text{DWconv}_{5\times5}(F)\right) + \text{DWconv}_{3\times3}(F) \tag{7}$$

where $P_{\text{avg}}$ represents average pooling operation. $\text{DWconv}_d$ and DWconv denote dilated depth-wise convolution and depth-wise convolution, respectively. Furthermore, we adopt a bifurcated gate unit structure to combine the LKA module with the MPK module. This allows us to establish interdependencies between distant pixels, enhancing the network's ability to capture long-range dependencies. Simultaneously, we retain central features through similar residual structures to avoid feature sparseness, ensuring that the network maintains a balance between capturing distant relationships and preserving local details. The structure can be expressed as follows:

$$Z = \text{Output}_{\text{LKA}} \otimes \text{Output}_{\text{MPK}} \oplus \text{Output}_{\text{MPK}}. \tag{8}$$

Here, $X_{\text{atten}}$ is the output of the LKA module, which is normalized to the range of (0, 1) after sigmoid. $\text{Output}_{\text{MPK}}$ is the output of the MPK module, which is multiplied element-wise with the processed $X_{\text{atten}}$.

### E. FFN Module

The FFN module primarily operates in the channel dimension [44]. It transforms and maps input feature maps through a series of batch normalization and activation layers to extract and enhance feature representations. Furthermore, we integrate $3 \times 3$ depth-wise separable convolution in the traditional FFN to capture the spatial correlation between adjacent feature pixels. The purpose of the FFN module is to introduce nonlinear transformation and increase the diversity and richness of the features, thereby enabling the model to better capture abstract features and semantic information in SAR images. Through the FFN module, we learn more discriminative feature representations, which in turn improves the model's detection performance and accuracy in identifying ship targets.

The module's design leverages the channel-wise transformation and the spatial correlation between neighboring feature pixels to generate a more comprehensive and informative feature representation. By incorporating the depth-wise separable convolution, the FFN module can efficiently capture the spatial relationships between adjacent feature pixels, complementing the channel-wise transformations. This combination of channel-wise and spatial-wise processing allows the model to learn a more holistic and meaningful feature representation, ultimately enhancing the overall performance in ship target detection tasks.

### F. Edge Self-Attention Module

In SAR image ship detection, traditional feature extraction methods combined with frequency-domain information provide advantages in enhancing detection robustness and improving the distinction between targets and background. By incorporating frequency-domain information, the model can further suppress sea clutter, capture global features, and enhance the edge contrast of small targets, effectively reducing edge blurring and detail loss. Our proposed Edge self-attention mechanism aids the network in achieving stronger detection capabilities in complex scenes, resulting in more stable and accurate performance in dynamic environments. Using frequency domain information to suppress sea clutter has attracted widespread attention.

SARDet-100K [45] proposed a multistage filter enhancement pre-training framework combined with traditional feature extraction methods. In the input stage of the network,

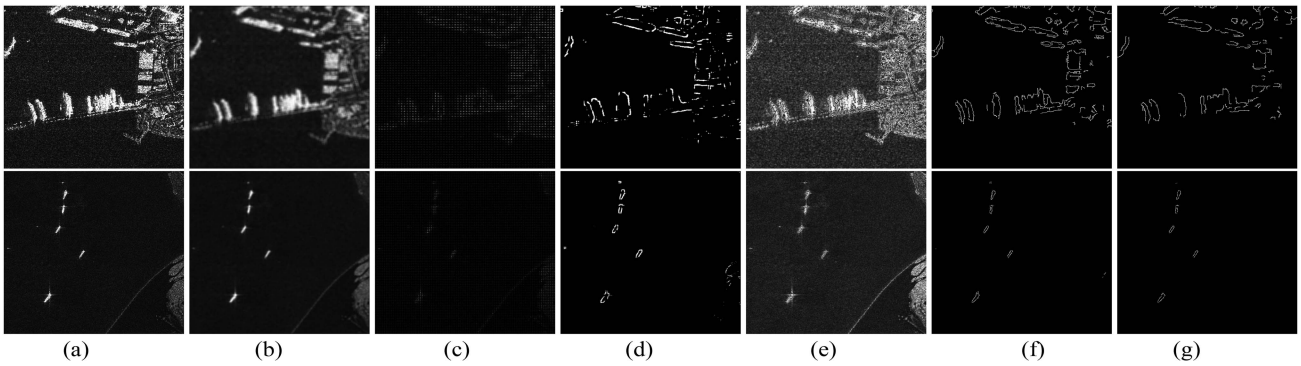



Fig. 6. Visualization of handcrafted features on SAR images. (a)–(g) The Original Image, Haar, HOG, GRE, WST, Canny, WST+Canny.

traditional manual feature descriptors and original pixels are combined as enhanced information of the original information. Feng et al. [46] proposed a SAR unsupervised interference detection model combining Canny edge detection and visual transformation (CEVIT). CEVIT takes the time-frequency spectrum as input, and realizes interference detection in complex interference environments with multiple interferences and multiple interference types through feature extraction modules and detection head modules. Wang et al. [47] designed a BD-FPN to obtain the multiscale features of the ship target and fuse the semantic features extracted from the spatial domain with the textural and detail features extracted from the frequency domain.

In SAR image ship detection, traditional feature extraction methods combined with frequency-domain information provide advantages in enhancing detection robustness and improving the distinction between targets and background [48]. Traditional feature extraction methods offer efficient computation and good interpretability, allowing for the extraction of key information, such as edges, textures, and shapes, even in complex backgrounds and under significant speckle noise. By incorporating frequency-domain information, the model can further suppress sea clutter, capture global features, and enhance the edge contrast of small targets, effectively reducing edge blurring and detail loss. Our proposed edge self-attention mechanism aids the network in achieving stronger detection capabilities in complex scenes, resulting in more stable and accurate performance in dynamic environments.

The main structure of the ship target in the SAR images is made of materials with high radar reflectivity [49]. Therefore, whether the ship target is at sea or on the shore, its edge will change greatly with the reflectivity of the surrounding sea surface, resulting in obvious edge features or gradient changes in the SAR images. Compared with visible light images, this feature is particularly obvious in SAR images. Therefore, this article innovatively applies the idea of obvious gradient changes around ships in SAR images to ship detection. We used a variety of traditional feature extraction methods to process the original image, including Haar, Hog, GRE, WST, Canny, and WST and Canny combined. As shown in Fig. 6, this module can efficiently extract the edge information of the ship target in the image, while effectively removing the interference of the land background. Finally, we chose Wavelet Transformer and Canny as the traditional feature extraction methods for our model. In order to enhance the detection network's ability to focus on the edges of objects, we designed the Edge self-attention architecture shown in Fig. 3. First, the original image is denoised using wavelet transform [50]. The SAR image is processed using the Canny operator to obtain the gradient matrix of the corresponding pixels in the image. Finally, the gradient matrix corresponding to the original image is subjected to maximum pooling operation to obtain multiple gradient matrices with the same size as the feature map output by each layer of the backbone network. We assume that the input image is X, and the specific processing flow is as follows:

$$E_0 = \text{Canny}\left(\text{WaveletDenoising}(X)\right). \tag{9}$$

Then, perform multiple rounds of maximum pooling on E0 to obtain [E1, E2, E3, E4], which is the same size as the feature map [F1, F2, F3, F4] of each stage of the backbone. In the edge attention mechanism, the input feature and the gradient information image of the target are first fused to calculate the 2-D gradient. The obtained gradient matrix is normalized by sigmoid and used as the weighted value of the corresponding pixel in the input feature. Finally, the feature map enhanced by edge attention is used for top-down FPN.

This module enhances the contour information of ship targets with obvious gradient in features. It aggregates and enhances the rich contour information in different layers, improving the contrast between the target and the noisy background.

## III. Implementation Details

This section presents the experimental results of the proposed method on two commonly used SAR images ship detection datasets. First, we introduce the SSDD and RSDD-SAR datasets and the parameter settings of the experiments. Then, the evaluation metrics used are described.

### A. Datasets and Settings

We utilize two datasets, SSDD [51] and RSDD-SAR [52], to evaluate the effectiveness of our proposed approach. SSDD is the first public ship dataset designed for SAR images, containing 1160 images showing 2456 ships with consistent OBB annotations. Serving as a key benchmark in SAR image ship detection, SSDD helps in assessing the performance of our method. On the other hand, RSDD-SAR, introduced in 2022,

is a dataset for SAR ship rotating frame detection consisting of 7000 multiresolution slices containing 10 263 ship instances. The SSDD dataset comprises slightly over 1000 images, and there is a significant difference in data volume between the two datasets, enabling a more comprehensive evaluation of the algorithm's generalization performance. In addition, the RSDD-SAR dataset includes a majority of small target ships, accounting for 60% of the total number, allowing for a thorough evaluation of the model's detection capability. The OBB annotations of two datasets are both annotated in DOTA [53] format.

For both the SSDD and RSDD-SAR datasets, we set the input size to be 608×608 and 512×512, respectively. Training is conducted over 120 epochs for SSDD and 72 epochs for RSDD-SAR. Our models are optimized using the AdamW optimizer [54], with an initial learning rate of 0.0002, a weight decay factor of 0.05, and a momentum of 0.9.

The models are implemented within the MMRotate [55] framework, built on the PyTorch deep learning framework, and operating on Ubuntu 22.04. Model training is conducted using 2 RTX3090 GPUs with a batch size of 8, while testing utilizes a single RTX3090 GPU.

### B. Evaluation Metrics

Some commonly used evaluation criteria are employed to assess the accuracy and completeness of the ship detection task: Recall, precision, AP50, AP75, and F1 score. In addition, we used floating point operations (FLOPs) and Params to measure the space complexity of the model.

Precision (P) represents accuracy, which is the ratio of correctly detected ship samples to all predicted positive ship samples. Recall (R) represents the recall rate, which is the ratio of correctly detected ship samples to all labeled positive ship samples. The mAP represents the average value of each category of AP, which is the area enclosed by the PR curve. When there is only one class, the mAP is equivalent to AP. AP50 represents the average precision at an IoU threshold of 0.5, aligning with the evaluation metric used in Pascal VOC. AP75 represents the average precision at an IoU threshold of 0.75. The calculation formulas for several indicators are as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{10}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{AP} = \int_0^1 \mathrm{P}(\mathrm{R})\,\mathrm{dR} \tag{12}$$

where TP represents true positives, referring to the correctly detected positive samples. FP represents false positives, indicating incorrectly detected positive samples. FN represents false negatives, denoting the incorrectly detected negative samples [56]. In addition, the F1 score is introduced to comprehensively evaluate the accuracy and completeness of the model

$$F1 = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{13}$$

TABLE I
IMPACT OF DIFFERENT NUMBER OF LK-BLOCKS AT EACH STAGE

| Blocks | R(%) | P(%) | AP50(%) | AP75(%) | Params(M) | FLOPs(G) |
|---|---|---|---|---|---|---|
| (2, 2, 4, 2) | 97.1 | 89.7 | 96.5 | 66.0 | **8.85** | **10.14** |
| (2, 2, 6, 2) | 97.3 | **91.4** | **97.0** | **67.7** | 9.63 | 11.31 |
| (3, 3, 9, 3) | **97.6** | 88.2 | 96.3 | 66.1 | 12.80 | 13.97 |
| (4, 12, 20, 4) | 96.5 | 89.3 | 96.1 | 63.2 | 20.33 | 31.97 |

TABLE II
IMPACT OF DIFFERENT MULTIPLE STAGES USING LARGE CONVOLUTION KERNEL

| Stage | SSDD | | | RSDD-SAR | | |
|---|---|---|---|---|---|---|
| | R(%) | AP50(%) | AP75(%) | R(%) | AP50(%) | AP75(%) |
| 1, 2 | 95.8 | 95.5 | 66.1 | 93.7 | 92.9 | 63.7 |
| 2, 3 | **97.4** | **97.2** | **67.5** | **94.2** | **93.8** | **64.3** |
| 3, 4 | 95.6 | 95.4 | 66.6 | **94.2** | 93.5 | 64.2 |
| 1, 2, 3, 4 | 96.8 | 96.4 | 67.1 | 93.9 | 93.2 | 64.0 |

## IV. EXPERIMENTS

In this section, we conducte experiments on SSDD and RSDD-SAR datasets to demonstrate the effectiveness of LKE-Det. First, we verify the effectiveness of each module of the proposed method through ablation studies. Then, we compare the proposed LKE-Det with other state-of-the-art optical-based and SAR image-based oriented object detectors on multiple evaluation metrics. The bold values in Tables represent the best performance among all methods.

### A. Ablation Experiments

A series of ablation experiments were conducted on the SSDD dataset to assess the contribution of each design component and the overall performance of the proposed method. All experiments were carried out within the Oriented R-CNN framework, and the performance was evaluated by multiple metrics, including P, R, F1 score, and AP. Initially, the effectiveness of the proposed modules, the impact of different numbers of LK-blocks at various stages, and the influence of the large convolution kernel in the different stage of LKA module were analyzed. Subsequently, the proposed modules were gradually integrated into the original network, and the effects of each module and the overall network were compared. Finally, for better illustration, we visualize the effect of each module in feature maps.

*1) Backbone of LKE-Det Analysis:* The proposed backbone is segmented into four stages, each containing a varying number of LK-blocks. The adjustment of block numbers in each stage is explored to assess its impact on experimental outcomes. According to Table I, optimal detect results are obtained with N = (2, 2, 6, 2), while maintaining few network parameters and computational complexity.

Based on the above experimental results, we speculate that it may not be necessary to use large kernel convolution in all stages. We validated the effectiveness of using large convolution kernels only in a few specific stages. The outcomes are detailed in Table II. Notably, employing large kernel convolution solely

TABLE III
IMPACT OF EACH COMPONENT OF LKE-DET

| | ResNet-50 | LKA | MPK | Edge-FPN | R(%) | P(%) | AP50(%) | AP75(%) | FLOPs (G) | Params (M) | FPS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (1) | ✓ | | | | 94.7 | 93.1 | 93.8 | 62.8 | **47.85** | 41.13 | 14.9 |
| (2) | | ✓ | | | 95.6 | 95.2 | 95.8 | 66.4 | 60.56 | **31.0** | **16.7** |
| (3) | | ✓ | ✓ | | **97.8** | 94.8 | 96.6 | 67.5 | 71.75 | 33.4 | 14.5 |
| (4) | ✓ | | | ✓ | **96.8** | 95.8 | 96.2 | 66.8 | 52.1 | 43.13 | 13.8 |
| (5) | | ✓ | ✓ | ✓ | 97.4 | **96.0** | **97.2** | **67.7** | 73.79 | 33.53 | 15.2 |

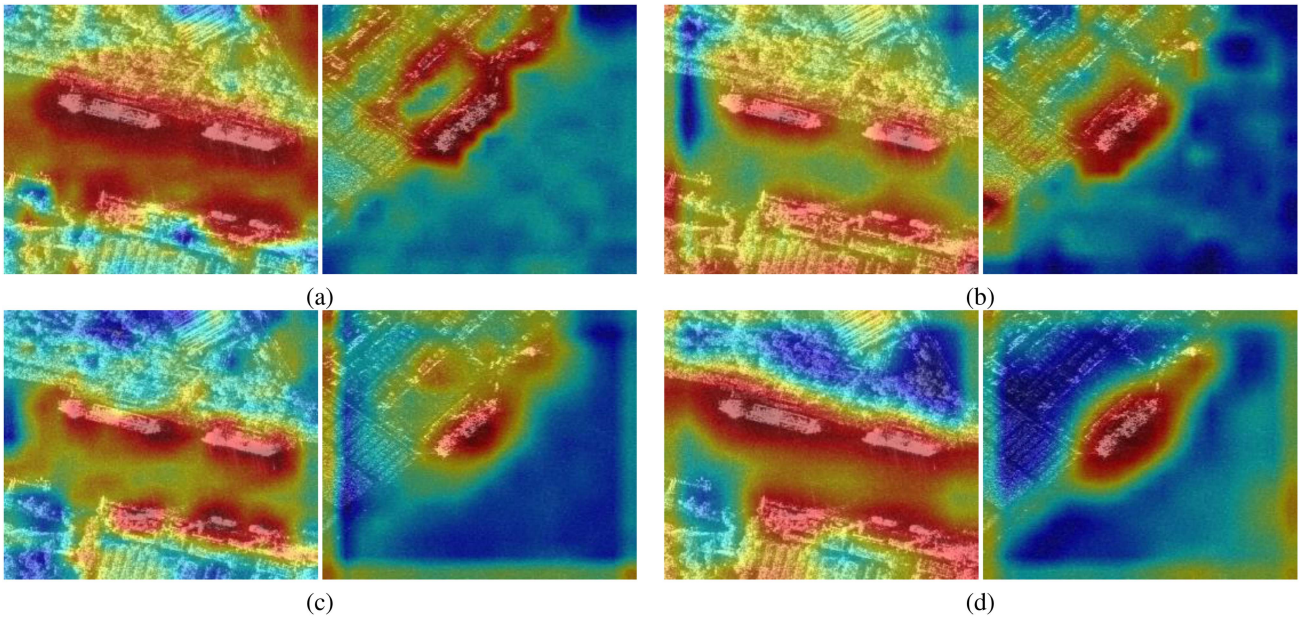



Fig. 7. Visualization of the feature maps extracted by Baseline and Our Method. (a) Feature maps of Resnet-50. (b) The output after backbone (Resnet-50) and FPN. (c) Feature maps of the backbone of LKE-Det. (d) The output after backbone(LK-blocks) and Edge-FPN.

in the second and third stages yields the best detection performance and minimizes missed detections. Table II shows that using larger kernels in the first stage will reduce the accuracy. We argue that having large kernels in the lower stages causes the low-level features to no longer be confined to small local areas. We think when the ship targets in SAR images are small, large convolution kernels with dilated rates may weaken the extraction of small target features. For smaller-scale ship targets, richer local low-level features are require. So we employ large convolution kernel in the middle-level stages.

*2) Proposed Modules Analysis:* Under the architecture with Oriented R-CNN as the baseline, we replace the proposed modules into the original network. Table III shows the overall results of the ablation experiment, which shows that each module we proposed significantly improves the detection performance. Compared with the baseline, our detector achieves significant improvement in $AP$ at different $IoU$ thresholds. Specifically, it improves by 3.4% and 4.9% on $AP_{50}$ and $AP_{75}$, respectively. In addition, the recall rate and precision rate are improved by 2.7% and 2.9%, respectively. These results clearly show that the proposed components improve the localization accuracy of ship objects and reduce false positives.

*3) Feature Map Visualization for Multiple Structures:* To better demonstrate the effectiveness of LK-Blocks and Edge self-attention module, we visualize the output feature maps of the original backbone (Resnet-50) [57], the original FPN, the backbone of our method, and the FPN equipped with Edge self-attention mechanism. For the output feature maps, we consider the maximum value in the feature matrix to be the most important and visualize it. As shown in Fig. 7(a) and 7(b), since the scattering characteristics of some buildings on land are similar to those of ship targets, the original network has a high attention to these buildings. This makes it difficult for the detector to detect ships with complex background interference, which may lead to inaccurate positioning and false detection. To address these problems, our proposed LK-Block effectively avoids the network's attention to land buildings by making full use of contextual information. Although land buildings also have strong scattering characteristics, the change gradient of their surrounding pixels is far less obvious than that of ship targets. Based on the characteristics of SAR images, our proposed edge self-attention mechanism can effectively enhance the detector's attention to ship targets while reducing attention to land buildings and ocean areas. In Fig. 7(c), the ship target becomes

TABLE IV
COMPARISON WITH STATE-OF-THE-ART ORIENTED DETECTION ALGORITHMS IN REMOTE SENSING IMAGES ON SSDD AND RSDD-SAR

| Method | SSDD | | | | | | RSDD-SAR | | | | | | Params(M) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | R(%) | P(%) | F1(%) | AP50(%) | AP75(%) | FPS | R(%) | P(%) | F1(%) | AP50(%) | AP75(%) | FPS | |
| R3Det | 91.0 | 90.0 | 90.49 | 90.7 | 49.5 | **14.2** | 89.1 | 87.1 | 88.09 | 87.7 | 48.2 | **10.2** | 41.58 |
| S2A-Net | 89.4 | 88.8 | 89.09 | 89.5 | 40.4 | 14.8 | 90.1 | 89.7 | 89.9 | 88.8 | 44.9 | 14.4 | 38.54 |
| RoI Transformer | 92.3 | 90.8 | 91.54 | 91.8 | 51.4 | 16.9 | 91.6 | 88.5 | 90.02 | 90.6 | 51.2 | 15.8 | 55.03 |
| Oriented RepPoints | 88.5 | 84.3 | 85.28 | 82.9 | 41.6 | 15.5 | 89.6 | 80.5 | 84.8 | 84.9 | 30.1 | 18.9 | 36.6 |
| Oriented Faster R-CNN | 91.0 | 92.0 | 91.49 | 90.0 | 42.4 | 19.3 | 89.8 | 84.3 | 86.9 | 87.3 | 31.8 | 19.3 | 41.12 |
| Oriented R-CNN | 94.7 | 93.1 | 93.89 | 93.8 | 62.8 | 14.9 | 92.2 | 90.8 | 91.5 | 91.8 | 59.7 | 13.6 | 41.13 |
| Oriented FCOS | 92.5 | 92.3 | 92.40 | 91.9 | 52.9 | 18.3 | 91.5 | 90.3 | 90.9 | 89.4 | 52.3 | 15.6 | **31.89** |
| Our method | **97.6** | **96.0** | **96.9** | **97.4** | **67.7** | 15.2 | **94.2** | **92.8** | **93.5** | **93.8** | **64.3** | 14.8 | 33.53 |

more prominent, and the global context captured by LK-Blocks significantly suppresses the activation value of the building. At the same time, as shown in the Fig. 7(d), FPN equipped with edge self-attention mechanism can further solve the above problems and effectively improve the results of accuracy and recall rate.

Overall, multiple ablation experiments have verified that each module in LKE-Det is beneficial for SAR ship detection, and our method achieves the best balance between all indicators.

### B. Comparative Experimental Results and Analysis

To confirm the feasibility and generalization ability of our method, we conduct experiments on two datasets of SSDD and RSDD-SAR and compare it with existing state-of-the-art methods. This includes a series of comparative experiments with the most advanced oriented target detection algorithms in the fields of remote sensing images and SAR images.

*1) Comparison With the State-of-The-Art Algorithms:* We compare our method with seven methods including R3Det [26], S2ANet [25], RoI Transformer [24], Oriented Faster R-CNN [28], Oriented RepPoints [58], Oriented FCOS [59] and Oriented R-CNN [28], which are all state-of-the-art oriented object detectors in remote sensing. The results on SSDD dataset and RSDD-SAR dataset are listed in Table IV. Our method outperforms all comparison methods, achieving superior detection results. As can be seen from Table IV, the proposed method achieves 97.4% and 93.8% AP50, 67.7% and 64.3% AP75 on SSDD and RSDD-SAR, respectively, and outperforms all other methods in terms of both precision and recall. AP75 is greatly improved on both datasets, which confirms that the proposed method can more accurately locate the ships in SAR images, and the bounding box contains less background, which is conducive to the recognition of the ships. The highest P and R indicate that our method can capture more positive samples and fewer negative samples, which better aligns with the genuine requirements of object detection. Moreover, our method has significantly improved the detection accuracy, while ensuring the detection speed and model scale of the detector.

To comprehensively compare the performance of different methods, four representative methods are chosen for visualizing the detection results: R3det [26], RoI Transformer [24], Oriented R-CNN [28], and Oriented FCOS [59]. These methods almost cover the most representative methods in the field of oriented object detection. The detection results of five methods, including LKE-Det, on the SSDD and RSDD-SAR datasets are shown in Figs. 8 and 9. Various complex scenes are selected to compare the ship detection performance of the methods, including densely arranged ships, ships docked at the shore and sparely distributed small ships. In the images, the green box indicates the detection box, the blue box represents the undetected target, and the yellow box signifies the false alarm. In the SSDD dataset, as shown in the first, second, and fourth rows of Fig. 8, for the case where ships are densely docked adjacent to each other, the edges of the ship targets are unclear and are interfered by the complex background of the shore. Our method demonstrates higher accuracy in predicting ship positions with fewer missed and false alarms, as evident in the figure. In addition, in the third row of Fig. 8, for small-scale ship targets with unclear shape features, our method misses fewer ships. In RSDD-SAR dataset, the first and second rows in Fig. 9 shows a harbor scene where ships of different sizes are densely docked. From the scene, it can be seen that most algorithms only detect ship targets with obvious features, while our algorithm can detect more small ships arranged with unclear boundaries.

The edge self-attention mechanism does increase the computational load and reduce the processing speed. Although we observed that adding the edge self-attention module affects the inference speed due to the increased computational complexity, the impact is not drastic. This is because we process images using a single channel, and both the Canny and wavelet algorithms use higher thresholds, which simplifies the processing flow. Importantly, the addition of the edge self-attention module enhances edge feature representation and overall accuracy. In addition, the inference speed of our customized LK Block module is significantly better than that of backbone networks such as ResNet-50 and ResNet-101. Compared with ResNet, the number of channels in each stage of LK-Blocks is reduced by half. Although LK-Block uses larger convolution kernels, it is better than ResNet in terms of inference speed and parameter amount, and achieves higher detection accuracy. Our proposed method achieves excellent detection accuracy while the inference speed is still competitive.

Overall, our method has better detection performance than other popular remote sensing oriented target detection methods in the field of ship target detection in SAR images.

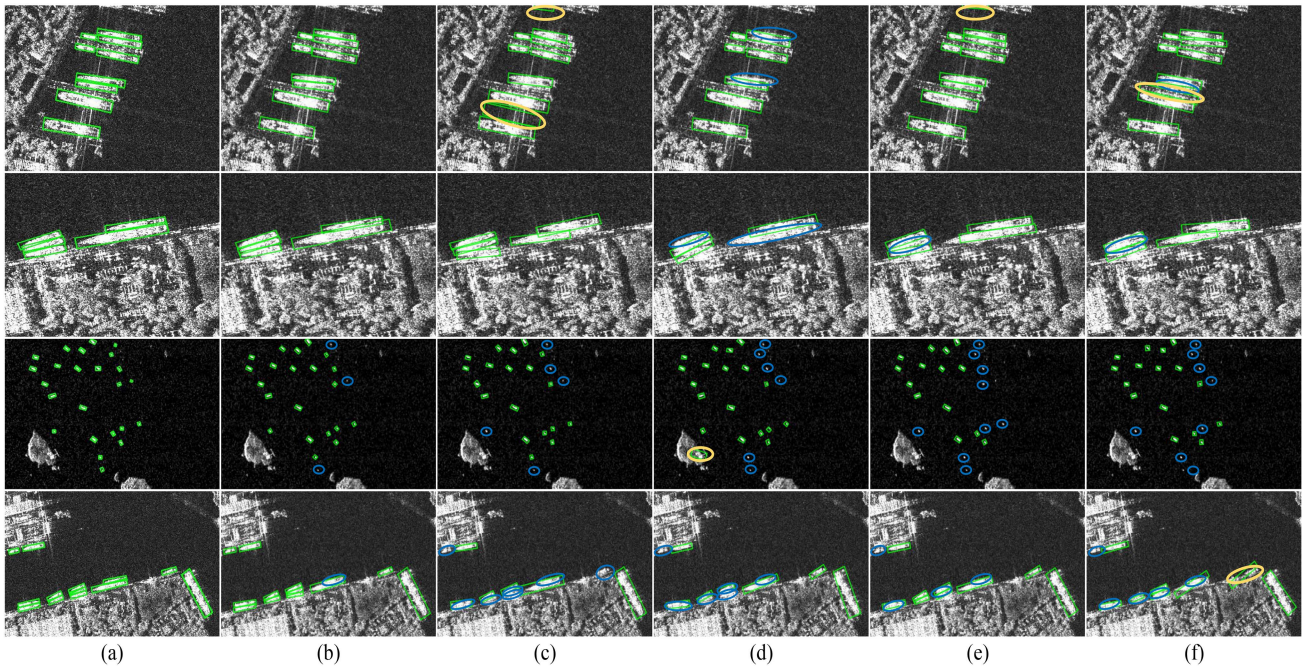



Fig. 8. Visualization of detection results of different methods on the SSDD. Green boxes represent the detection results, blue boxes represent the missed ships and the yellow boxes represent the false alarms. (a) Ground truth. (b) Ours. (c) R3Det. (d) RoI Transformer. (e) Oriented R-CNN. (f) Oriented FCOS.

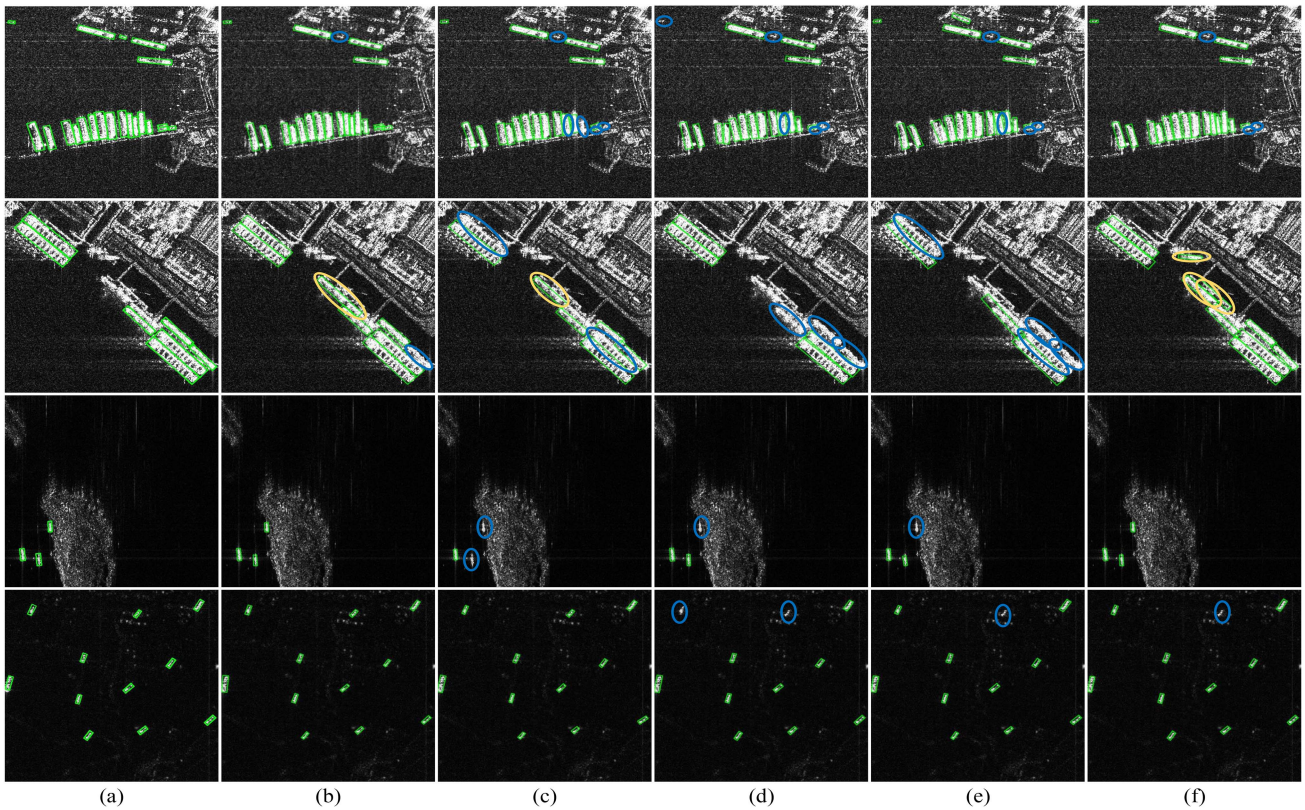



Fig. 9. Visualization of detection results of different methods on the RSDD-SAR. Green boxes represent the detection results, blue boxes represent the missed ships and the yellow boxes represent the false alarms. (a) Ground truth. (b) Ours. (c) R3Det. (d) Roi Transformer. (e) Oriented R-CNN. (f) Oriented FCOS.

TABLE V
COMPARISON WITH THE SAR TARGET DETECTION ALGORITHMS

| Method | R(%) | P(%) | F1(%) | Ap50(%) | Ap75(%) | Params(M) |
|---|---|---|---|---|---|---|
| GFB-Net [60] | 85.6 | 94.0 | 89.6 | 93.0 | – | 41.4 |
| OFF-Net [61] | 93.9 | 92.0 | 92.9 | 95.0 | 51.7 | – |
| SRT-Net [62] | 95.8 | 95.1 | 95.7 | 94.7 | 39.3 | – |
| FADet [63] | 94.5 | 94.7 | 94.6 | 91.0 | – | 42.8 |
| FbR-Net [23] | 92.7 | 94.0 | 93.4 | 94.10 | 59.0 | **32.5** |
| BTRDet [64] | 93.2 | – | – | 90.31 | 42.25 | – |
| DRPSM [35] | – | **96.2** | – | 90.7 | **67.8** | 43.3 |
| Ours | **97.6** | 96.0 | **96.7** | **97.4** | 67.7 | 33.53 |

*2) Comparison With the SAR Target Detection Algorithms Based on Deep Learning:* To further verify the performance of LKE-Det, we conducted further comparative experiments with the most advanced oriented detection algorithms based on SAR images. In SSDD dataset, we compared six algorithms: GFB-Net [60], OFF-Net [61], SRT-Net [62], FAdet [63], FbR-Net [23], BTRDet [64], and DRPSM [35]. As shown in Table V, the precision of our method is 1.8% higher than 95.8% of the highest SRT-Net, the F1 score is 1.0% higher than SRT-Net and the AP50 is also higher than other algorithms. From Params in the table, FbR-Net achieves the least parameters, but ours is only slightly more, and the accuracy of our network is significantly better than FbR-Net. From the comprehensive evaluation of multiple indicators, our method achieves a relatively advanced detection performance. In general, the proposed method achieves a better balance between detection performance and network scale, and has great advantages in detection performance and anti-interference ability in SAR images.

## V. CONCLUSION

This article proposes a two-stage detector for oriented ship detection in SAR images, utilizing large convolution kernels and edge self-attention. SAR images have unique prior knowledge as a form of remote sensing data. Therefore, leveraging the contextual semantic information of ships can significantly enhance the detector's performance. Considering the limited receptive field of traditional small kernel convolution, we adopt the concept of large convolution kernel decomposition to broaden the effective receptive field and capture remote contextual information. Simultaneously, we employ multiple parallel multiscale convolutions to extract object features of varying scales and capture local contextual feature information. Furthermore, ships in SAR images exhibit stronger scattering characteristics than the surrounding environment, with noticeable gradient changes around them. Building on this innovative concept, we design a feature pyramid structure integrated with an edge self-attention mechanism, which effectively highlights the characteristics of ships while reducing the interference of complex backgrounds, particularly for small ships in SAR images. Extensive comparative experiments conducted on SSDD and RSDD-SAR datasets demonstrate that our method surpasses other popular oriented target remote sensing detectors and most state-of-the-art SAR target detection algorithms.

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